KAVANA P

Bengaluru, India | kavanapcse1@gmail.com | +91 7892557553 | linkedin.com/in/kavana-padmakumar-1a457b393 | github.com/Kavana-Padmakumar

OBJECTIVE

Third-year Computer Science Engineering student working toward a career in machine learning research. Hands-on experience in self-supervised representation learning, applied reinforcement learning, mechanistic interpretability, and on-device time-series modeling. Comfortable building ML systems end to end, from architecture through a deployed interface, with a track record of shipped, award-winning projects.

EDUCATION

B.E., Computer Science Engineering *2024 - 2028 (Expected)*

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Data Science, Data Structures & Algorithms

TECHNICAL SKILLS

Machine Learning & Deep Learning: PyTorch, Self-Supervised Learning, Representation Learning, Time-Series Modeling, Computer Vision, Reinforcement Learning, LSTMs, Transformers

Research Methods: Activation Patching, Centered Kernel Alignment (CKA), Statistical Analysis (bootstrap, mixed-effects models)

Engineering: Python, FastAPI, Docker, Git, PostgreSQL, Next.js, TypeScript, React

Applied / Supporting: ESP32, Sensor Fusion, Embedded C++

CERTIFICATIONS

- Nation SkillUp Program — Python, Data Analytics, AI Tools & Git — GeeksforGeeks, Jan 2026
- Data Science Skill-Up — Pursuit Technologies
- 3D Drone Making

EXPERIENCE

Data Science Intern — Pursuit Future Technologies *Jan 2026 - Apr 2026*

- Applied Python, machine learning, data preprocessing, and exploratory data analysis across applied predictive modeling projects; built a Customer Churn Prediction model and a Blind Auction application
- Received the Performance of Excellence award in recognition of internship work

RESEARCH PROJECTS

LOCUS - Causal Decomposition of Test-Time Adaptation *Solo Research | In Progress*

- Designed a causal framework to test whether test-time adaptation methods (TENT, BN-Adapt, SAR) improve accuracy via genuine representation change or decision-boundary shift alone
- Built a capacity-matched decision-boundary-only control model; applied activation patching and Centered Kernel Alignment (CKA) to attribute accuracy gains to specific network layers
- Developing an open-source diagnostic toolkit for analyzing test-time adaptation methods

PulseFrame - Self-Supervised Representation Learning for Physiological Time-Series *Solo Research | In Progress*

- Converted multivariate physiological signals (ECG, PPG, accelerometer) into 2D representations (Gramian Angular Fields, recurrence plots, scalograms) for vision-based modeling
- Pretrained a Vision Transformer with self-supervised objectives (contrastive / masked-image-modeling) on public wearable datasets (WESAD, PPG-DaLiA, MIT-BIH)
- Built a few-shot personalization pipeline adapting the pretrained model to a new individual from minutes of calibration data

PhysioShift - Personalized Physiological Signal Modeling *Team Project | In Progress*

- Team research initiative on physiological monitoring adapting to individual baselines rather than population averages; applications include VitalDrift (on-device cardiac early-warning system using a 14-day biosignal baseline) and Follicular (follicular-phase health tracking)

LakeGuard AI + AquaDrone Sentinel *Apr 2026*

- Built a real-time lake health monitoring dashboard paired with a floating ESP32 sensor drone (React, FastAPI)
- Placed 4th of 147 teams at the KSRSAC State-Level Student Innovation Exhibition, winning ₹15,000

ACHIEVEMENTS

- 4th Place of 147 teams, KSRSAC State-Level Student Innovation Exhibition (Govt. of Karnataka) for LakeGuard AI, winning ₹15,000
- 250+ problems solved on LeetCode, maintained through a sustained daily-practice streak across DSA topics
- 130-day streak on Duolingo
- Active hackathon participant across multiple Bengaluru-area college hackathons

LANGUAGES & INTERESTS

Languages: English, Kannada, Hindi, German, Japanese

Interests: AI-driven side projects, hackathons, travel, comics